

Survey paper

Deep learning-based depression recognition through facial expression: A systematic review

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ABSTRACT

Depression is a type of prevalent mental illness that can lead to suicidal or self-harm behaviors in severe cases. Recently, depression recognition has garnered extensive attention from the deep learning community due to its urgent need to assist conventional diagnostic methods. Deep learning-based depression recognition through facial expression (DL-FEDR) is one of the most popular research directions. Therefore, this paper tries to summarize advances from 2017 to 2024 in DL-FEDR. We focus on (1) Various approaches of DL-FEDR are divided into two categories: spatial and spatial-temporal features. (2) These methods are analyzed from metrics results, ethical privacy, application scenarios and technological advancements. (3) Present challenges and future directions of DL-FEDR systems are discussed.

1. Introduction

Depression is a common mental illness in the world. By 2030, approximately 350 million individuals worldwide will be affected by this mental disorder, according to the World Health Organization (WHO) [1]. This increase is expected to profoundly impact individual well-being, family dynamics, and societal structures [2,3]. Given the growing number of depression sufferers, there is an urgent need for early and effective identification, particularly in milder cases, to reduce suffering and facilitate timely, cost-effective treatment. Current diagnostic methods are mainly clinical interviews or questionnaires, which rely on the professional knowledge and experience of doctors and the active cooperation of patients. The limitations have prompted researchers to explore alternative approaches. In recent years, deep learning has shown promise in addressing this challenge. Studies have explored various modalities, including facial expression [4], text [5,6], speech [7], and EEG [8], among others [9–13]. Research has identified significant differences in facial expressions, eye contact, and other visual cues between individuals with depression and those without [14].

So far, facial expression analysis has emerged as a prominent approach in deep learning for diagnosing depression. This paper aims to provide a comprehensive review, focusing on the advancements in deep learning-based methods for depression recognition through facial expressions. It covers the period from 2017 to 2024, examining depression datasets, AVEC2013 [15], AVEC2014 [16] and D-vlog [17],

summarizing and reflecting on past and current research, and offering insights into future directions.

Our selected papers, which form the basis of this review, are predominantly sourced from prestigious journals and conference proceedings, including but not limited to the IEEE Transactions on Affective Computing, Expert Systems with Applications, and the IEEE Transactions on Circuits and Systems for Video Technology (see Table 1). This paper offers an in-depth exploration of deep learning-based depression recognition through facial expression (DL-FEDR), analyzing the progression of methodologies and identifying potential avenues for future research. The main contributions of this work are: (1) This review provides a comprehensive analysis of the DL-FEDR field from 2017 to 2024, categorizing the methodologies into spatial and spatial-temporal approaches. (2) The review examines these methods from various perspectives, including metric results, ethical privacy, application scenarios, and technological developments. (3) It also suggests open problems and future research directions, considering the broader impact and potential of DL-FEDR systems.

The structure of this paper is methodically laid out. Section 2 introduces the concept of depression and elaborates on the related datasets that are pivotal for the research within this domain. Section 3 lists specific approaches for DL-FEDR and categorizes them into spatial and spatial-temporal methodologies, further discussing these approaches concerning performance metrics, ethical privacy considerations, potential application scenarios, and technological advancements. Current

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0925-2312/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Table 1
Source Journals and Conferences.

Journal/Conference	Num
IEEE Transactions on Affective Computing	15
Expert Systems with Applications	3
ICASSP(IEEE International Conference on Acoustics, Speech and Signal Processing)	3
International Journal of Intelligent Systems	2
IEEE Transactions on Circuits and Systems for Video Technology	2
Neurocomputing	1
BIBM(IEEE International Conference on Bioinformatics and Biomedicine)	1
ICIP(IEEE International Conference on Image Processing)	1
CVPR(Computer Vision and Pattern Recognition)	1
Pattern Recognition Letters	1
IEEE Internet of Things Journal	1
IEEE International Conference on Automatic Face & Gesture Recognition	1
IEEE Transactions on Industrial Informatics	1
Neural Networks	1
IEEE Transactions on Knowledge and Data Engineering	1
AAAI Conference on Artificial Intelligence	1
IEEE Transactions on Neural Systems and Rehabilitation Engineering	1
Biomedical Signal Processing and Control	2
Digital Communications and Networks	1
Digital Signal Processing	1
Total	41

challenges and future prospects of DL-FEDR are described in Section 4. Finally, Section 5 summarizes the key findings and conclusions of this review.

2. Preliminaries

To understand DL-FEDR, we start by providing an overview of depression, including its definition, symptoms, types, causes, clinical diagnosis, and the importance of early treatment. Then, related depression datasets are introduced.

2.1. What is depression?

Depression, also known as major depressive disorder or clinical depression, is a prevalent and complex mood disorder that significantly impacts an individual's emotional state, thoughts, and behaviors [18]. It is characterized by a persistent feeling of sadness, loss of interest or pleasure in activities, and a range of emotional and physical problems [19] that can disrupt normal day-to-day activities and even lead to suicidal thoughts or attempts [20].

Symptoms of depression vary and can include feelings of sadness, tearfulness, emptiness, hopelessness, irritability, anger, sleep disturbances, fatigue, reduced appetite or weight changes, anxiety, agitation, slowed thinking or movements, feelings of worthlessness or guilt, difficulty concentrating, and unexplained physical problems.

Depression is classified into several types, including Major Depressive Disorder (MDD), Persistent Depressive Disorder (Dysthymia), Perinatal Depression, Premenstrual Dysphoric Disorder (PMDD), Depression with Psychotic Features, Seasonal Affective Disorder (SAD), and Bipolar Disorder [21]. Each type is characterized by specific symptoms and duration, with major depressive disorder involving symptoms for over two weeks that affect daily functioning.

The aetiology of depression is multifaceted, involving a combination of genetic [22], biological, environmental, and psychological factors. Genetic predisposition plays a role, with depression more common in individuals with family members who have the condition. Biological differences, such as brain chemistry and neurotransmitter function changes, are implicated in the disorder [23]. Hormonal changes, particularly in women, can trigger depressive episodes. Stressful life events, personality traits, and certain medical conditions also contribute to the development of depression.

Clinical diagnosis often relies on self-report scales and inventories (SRISs), with the Patient Health Questionnaire (PHQ-2/8/9) and Beck's

Depression Inventory (BDI) being commonly used [24]. The Hamilton Depression Rating Scale (HAMD) is an interview-based method contrasting with the self-assessment nature of the BDI. Both scales assign scores to rate the severity of depression, with HAMD focusing on neuro-vegetative symptoms and BDI on self-evaluations of negative symptoms [25].

The benefits of early treatment for depression are substantial. Seeking help at the first signs of depression can prevent the condition from worsening, reducing the risk of emotional, behavioral, and health complications. Early intervention can also improve treatment outcomes, allowing for a more effective response to medication, psychotherapy, or a combination of both.

2.2. Related depression datasets

DL-FEDR relies on substantial data to effectively train a discriminative model. However, due to depression's sensitive nature, particularly regarding the collection of facial videos, obtaining such data presents a challenge. Some datasets are private and less commonly accessible [26–28]. Therefore, we introduce depression datasets for public use: AVEC2013 [15] and AVEC2014 [16] and D-vlog [29]. Fig. 1 illustrates some examples from the AVEC2013 database, while Fig. 2 showcases image samples with occlusion, insufficient lighting, and variations in gaze direction.

The AVEC 2013 dataset is a subset of the Audio-Visual Depressive Language Corpus (AViD-Corpus), consisting of 340 video clips recorded from 292 subjects. Each clip features only one subject, with some individuals participating in multiple recordings. The duration of clips ranges from 20 to 50 min, with an average duration of 25, totaling 240 h of video. The subjects' average age is 31.5 years, with a standard deviation of 12.3 years, ranging from 18 to 63 years. In each clip, the subject is guided through various tasks, including sustained vowel pronunciation, loud speech during a task, counting from 1 to 10, reading excerpts from Max Frisch's novel "Homo Faber" and the German fable "Die Sonne und der Wind," singing nursery rhymes, recounting the best and saddest childhood memories, and telling imaginary stories, with guidance provided using PowerPoint (PPT). The AVEC 2013 depression dataset includes training, validation, and test sets containing 50 video clips.

The AVEC 2014 dataset is a subset of the AVEC 2013 database, focusing on two task categories selected from the original tasks to concentrate the analysis on emotion and depression. The selected tasks are (1) Northwind, where subjects read a passage from the fable "Die Sonne und der Wind" in German, and (2) Freeform, where participants

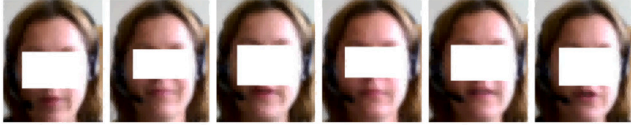
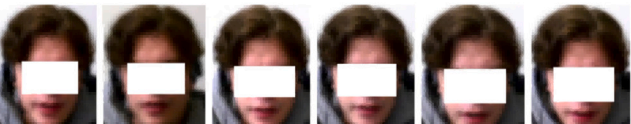
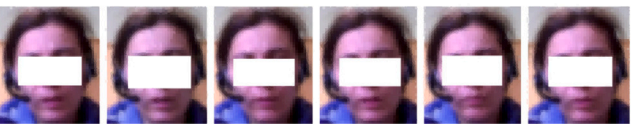
	BDI-II Score	Depression Level
	0	None(0-13)
	16	Mild(14-19)
	23	Moderate(20-28)
	34	Severe(29-63)

Fig. 1. Selected frames from subjects with varying BDI-II scores, illustrating different levels of depression [15]. To safeguard the individuals' privacy, we blurred the frames and obscured the eyes.

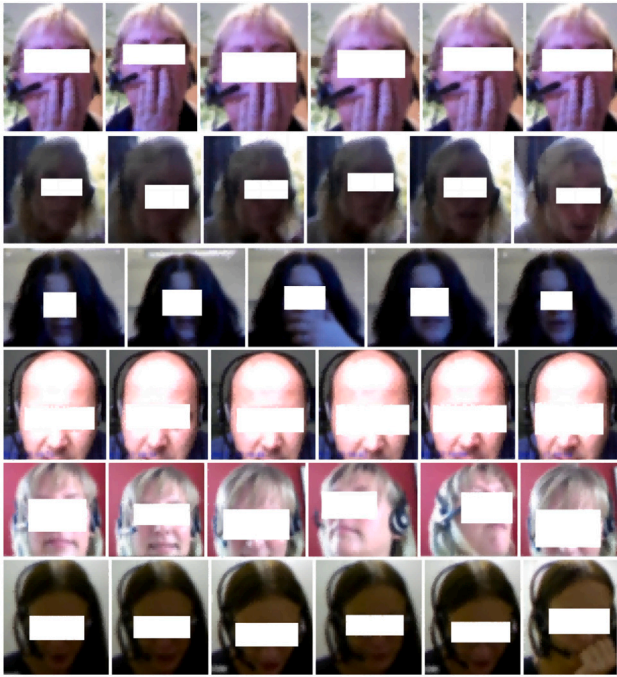


Fig. 2. Some image examples with occlusion, insufficient lighting, and looking up or down [15,16].

answer one of several questions in German, such as their favorite dish, best gift received, or a sad childhood memory.

In recent years, with the trend towards multimodality, an increasing number of studies have considered integrating facial expressions with other modalities, especially speech. The D-vlog is a multimodal collection of vlogs designed to detect depression by analyzing non-verbal behaviors. It comprises 961 vlogs from 816 users, totaling approximately 160 h of video content sourced from YouTube. To protect the facial privacy of users, the D-vlog only reveals two non-intuitive features: 68 facial landmarks extracted by the Dlib toolkit and acoustic features extracted by OpenSmile. Within the D-vlog dataset, there are

555 depression vlogs and 406 non-depression vlogs. The average length of depression vlogs is 640.12 s, while non-depression vlogs average 536.38 s. Additionally, there are twice as many females as males in the depression vlogs.

3. DL-FEDR

In this section, we introduce specific approaches for DL-FEDR. Based on how to process spatial and temporal information in videos, we divide the methods into two groups: spatial and spatial-temporal approaches for DL-FEDR(see Table 2).

3.1. Spatial features

The video can be viewed as a fusion of appearance and dynamic information. In the early stages of DL-FEDR, the focus was primarily on capturing spatial features in facial images without considering the temporal relationships within frames. Ten of the 41 studies reviewed emphasized using 2D-CNNs to extract spatial representations from facial images.

Zhou et al. [30] introduced DepressNet, which utilizes ResNet-50 with global average pooling to learn representations of depression and generate Depression Activation Maps (DAM) for identifying salient depression-related regions in input images. They also proposed a multi-region DepressNet model that jointly learns from various local face regions and full-face images, enhancing overall recognition performance (see Fig. 3).

In 2020, [31] proposed a Deep Residual Regression Convolutional Neural Network (DRR_DepressionNet) consisting of three main components: the C_M block, Resblock, and Global Average Pooling (GAP). This network employs image enhancement strategies, including histogram and adaptive equalization, to counteract lighting and head posture variations.

Similar to the previous methods [30,31], DepNet [32] employs a classic deep-learning architecture with innovative modifications. It integrates ResNet and SENet to create an enhanced model known as SE-ResNet-50. Utilizing pre-training and fine-tuning techniques, the network is adeptly trained to extract low-level features. Furthermore, DepNet employs a pivotal component called the Feature Aggregation Module (FAM) to distil high-level features (HLFs) indicative of depression.

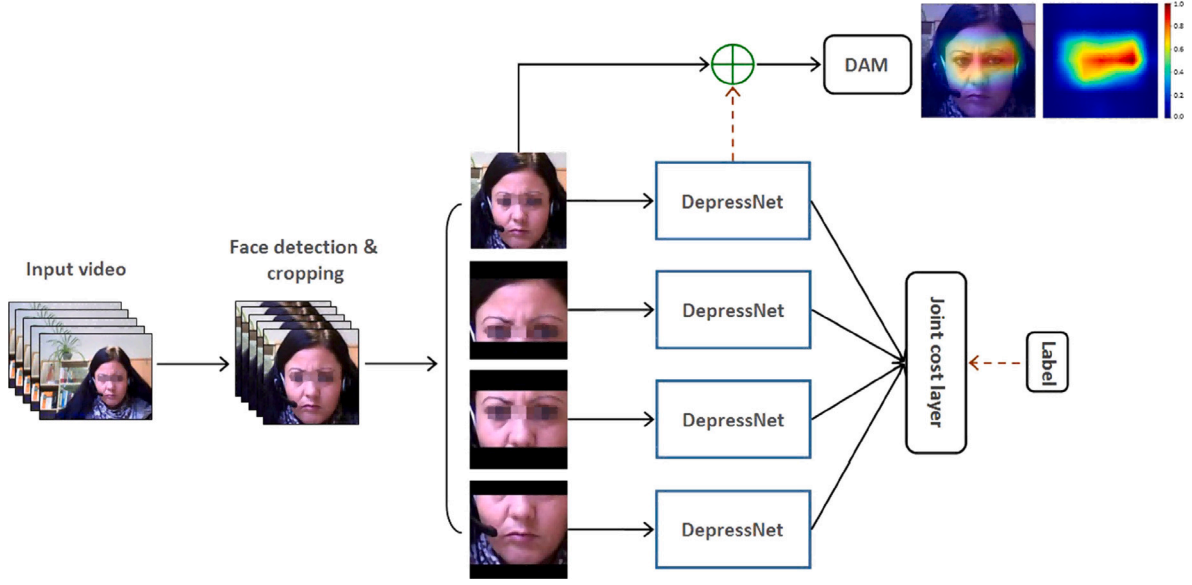


Fig. 3. Architecture of multi-region DepressNet [30].

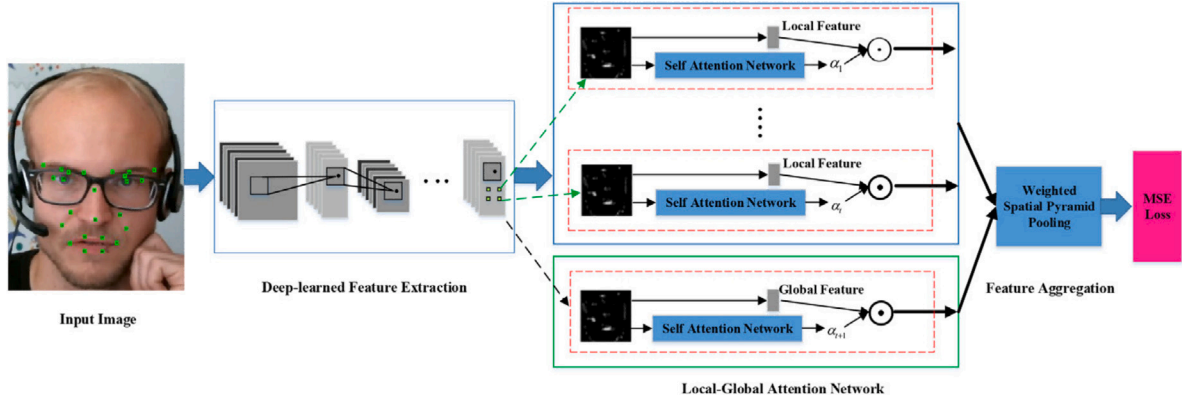


Fig. 4. Architecture of DLGA-CNN [33].

In 2021, He et al. [33] proposed a Deep Local Global Attention Convolutional Neural Network (DLGA-CNN) for depression recognition, which combines CNN with attention mechanism and weighted spatial pyramid pooling (WSPP). The framework includes two branches: Local Attention-based CNN (LA-CNN) for local patches and Global Attention-based CNN (GA-CNN) for global patterns. As depicted in Fig. 4, the Local-Global Attention-based CNN (LGA-CNN) is introduced to capture complementary information between the two branches.

Unlike previous approaches [30,31,33] employing regression with Euclidean loss, Melo et al. [34] introduce a deep learning architecture utilizing distribution learning to represent the ordinal relationship between facial images and depression levels. It employs an expectation loss function to estimate the underlying data distribution across depression levels, enhancing robustness to noisy and uncertain labeling.

Commonly utilized in facial depression recognition, approaches can be crafted manually or derived from deep learning models. Shang et al. [35] introduce the Local Quaternion and Global Deep Network (LQGDNet), which effectively integrates hand-crafted local with deep global features (refer to Fig. 5). The technique uses Quaternion XOR Asymmetrical Regional Local Gradient Coding (XOR-AR-LGC) to encode facial images in the quaternion domain, preserving color channel interdependencies for better feature extraction. The Quaternion Feature Extractor (QFE) combines this with quaternion-based Local Binary Patterns (LBP) to generate local quaternion features. A Local Deep Feature Extractor (LD FE) further refines features using deep convolutional

networks. Together, these form the Local Quaternion Representation Module (LQRM). On a larger scale, the Global Deep Representation Module (GDRM) utilizes ResNet-18 to extract global facial features. The LQGDNet merges LQRM and GDRM, combining local and global features via a multi-layer fully connected network to accurately predict depression scores.

He et al. [36] introduce an innovative approach known as the Self-Adaptation Network (SAN) for depression recognition. This method comprises four key modules: the Deep Feature Extraction Module (DFEM), which utilizes ResNet to extract informative features; the Self-Attention Module (SAM), which learns the importance of each image within a batch; the Square Ranking Regularization Module (SRRM), which partitions and regularizes weights; and the Re-Label Module (RM), responsible for re-labeling uncertain annotations. Consequently, this approach helps to address the challenge of noisy annotations in ADE tasks.

Both Zhang et al. [37] and Yu et al. [38] present innovative frameworks for automatic depression detection, emphasizing cloud-edge collaboration and deployability. Both frameworks leverage advanced deep learning models to improve detection using local and global features, optimizing accuracy and computational efficiency for practical applications. However, Zhang et al.'s focus on a two-stream network captures global and local features, emphasizing detail and security applications (see Fig. 6). In contrast, Yu et al.'s framework combines

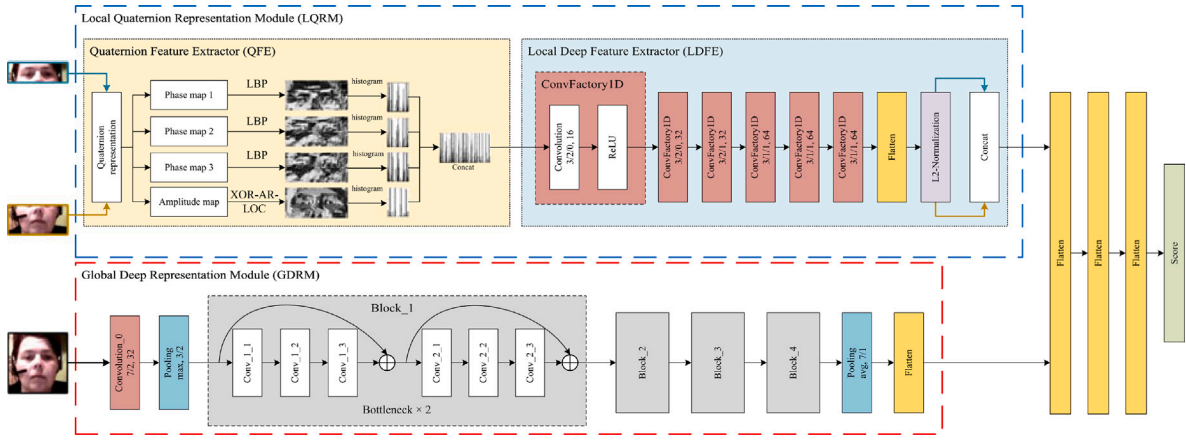


Fig. 5. LQGDNet for DL-FEDR [35]. The LQRM processes local color images, focusing on specific areas such as the mouth and eyes, while the GDRM takes the entire facial image as its input for analysis.

negative emotion screening with depression recognition, prioritizing efficiency and real-time analysis at the edge with cloud-based detailed analysis (refer to Fig. 7). Zhang et al. emphasize knowledge distillation for deployable models, while Yu et al. stress real-time capabilities and edge-cloud task off loading decisions.

3.1.1. Evaluation of spatial features for DL-FEDR: strengths and limitations

Spatial feature-based methods, typically utilizing 2D convolutional neural networks (CNNs), offer several advantages. Firstly, they have relatively low computational complexity, making them well-suited for deployment on resource-constrained devices. For example, lightweight models like EdgeER are used in edge servers for real-time emotion recognition with minimal computational cost [38]. Additionally, these methods are more privacy-friendly, as they can protect sensitive data by applying techniques like image blurring or de-identification, ensuring patient privacy [39].

However, the primary limitation of spatial feature-based methods is their inability to capture temporal information. These methods focus on individual frames, which means they overlook dynamic variations that unfold over time, such as changes in facial expressions or the continuity of movement. As noted by Melo et al. [40], relying solely on spatial features from single-frame images may fail to capture the nuanced and complex behavioral patterns of depression patients. This limitation can lead to suboptimal performance in recognizing complex emotional states, as the lack of temporal context impairs the model's ability to interpret evolving emotional behaviors. Consequently, these methods are less effective in handling dynamic, time-dependent expressions of emotions.

3.2. Spatial-temporal features

As DL-FEDR progressed, there was a shift towards spatial-temporal features within image frames. Among the 41 studies reviewed, 31 emphasized extracting spatial-temporal representations, capturing both facial appearance and dynamics.

Zhu et al. [41] propose a two-stream approach for predicting depression from videos. As illustrated in Fig. 8, both similar to the GoogLeNet [42], the appearance Deep Convolutional Neural Networks (DCNN) analyzes facial frames, while the dynamics DCNN captures facial movements using optical flow. The joint tuning layer combines appearance and dynamics to improve prediction accuracy.

Jazaery et al. [43] introduce a novel framework, RNN-C3D, for automatically diagnosing depression using video data. The framework leverages a 3D convolutional neural network (C3D) [44] to extract spatiotemporal features from different scales of facial regions and employs a Recurrent Neural Network (RNN) to further learn from the sequence

of these features. Two C3D models are trained: one on tightly aligned facial regions (C3D Tight-Face model) and another on more giant face regions (C3D Loose-Face model), to capture both facial expressions and head gestures.

Zhou et al. [45] introduce DJ-LDML for facial depression recognition, combining label distribution learning (LDL) and deep metric learning (DML). The C3D backbone extracts spatiotemporal features from videos, processed by a softmax layer for label distribution and a metric layer for discrimination. DJ-LDML exploits label relevance in depression data via LDL, augmenting training data per depression level without dataset expansion. DML focuses on learning a deep ordinal embedding using a label-aware histogram loss, maintaining semantic similarity for discriminative feature learning. The joint use of LDL and DML enhances spatiotemporal feature representation and discrimination, improving depression prediction.

Uddin et al. [46] propose a framework for estimating depression levels from video data using a two-stream deep spatiotemporal network. Spatial information is extracted using the Inception-ResNet-v2 network, and facial motions are captured using a volume local directional number (VLDN)-based dynamic feature descriptor. The VLDN feature map is fed into a convolutional neural network (CNN) for more discriminative features. Then, a multi-layer bidirectional long short-term memory (Bi-LSTM) model with temporal median pooling (TMP) integrates this temporal information (see Fig. 9).

Song et al. [47] introduce an innovative approach for automatic depression analysis using video-based human behavior primitives. The method employs OpenFace 2.0 [48] to extract a multi-channel time-series of facial action units (AUs), gaze directions, and head poses, which are then transformed into the frequency domain through Fourier Transform to capture multi-scale temporal dynamics of expressive behavior. To address variations in video lengths, the method proposes frequency alignment techniques, including zero-padding and frequency selection, to standardize the spectral representations. These representations are either visualized as spectral heatmaps or concatenated as spectral vectors, which are then fed into Convolutional Neural Networks (CNNs) and Artificial Neural Networks (ANNs) for depression analysis.

Niu et al. [49] present a multi-scale and multi-region facial dynamic representation method. In particular, it divides long-term videos into segments based on multiple time scales, capturing various facial regions' activities. Segment-level features are extracted using a 3D convolutional neural network (3D-CNN) to characterize dynamic changes across different durations and facial regions. Eigen evolution pooling (EEP) and gradient boosting decision tree (GBDT) techniques are then employed to aggregate these features, selecting discriminative elements for generating video-level features. Finally, support vector regression (SVR) is utilized to predict the depression level.

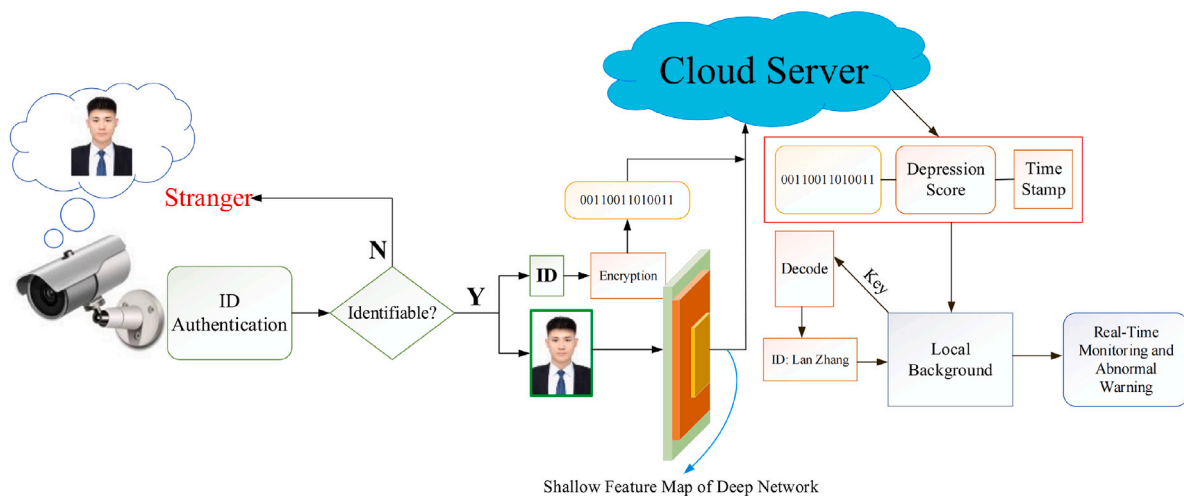


Fig. 6. A practical application scenario for remote depression recognition [37].

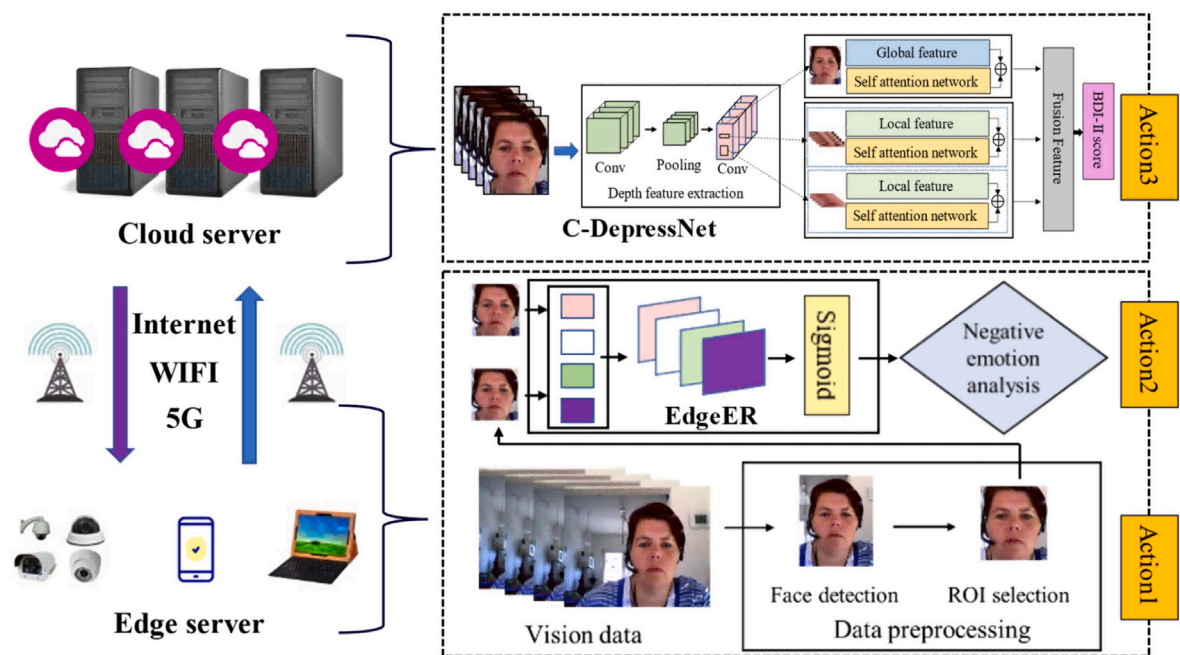


Fig. 7. The architecture of cloud-edge collaboration [38].

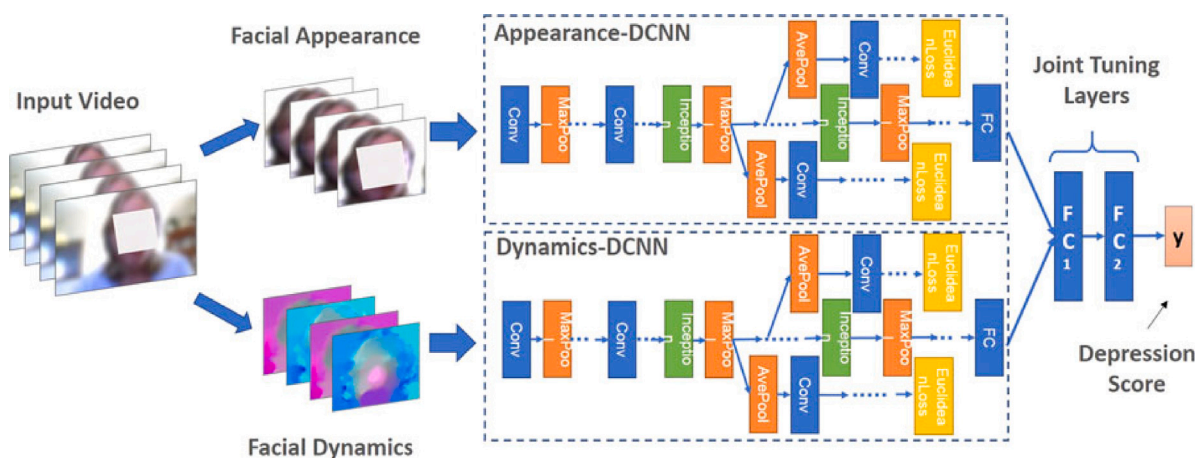


Fig. 8. Architecture of appearance and dynamics DCNNs [41].

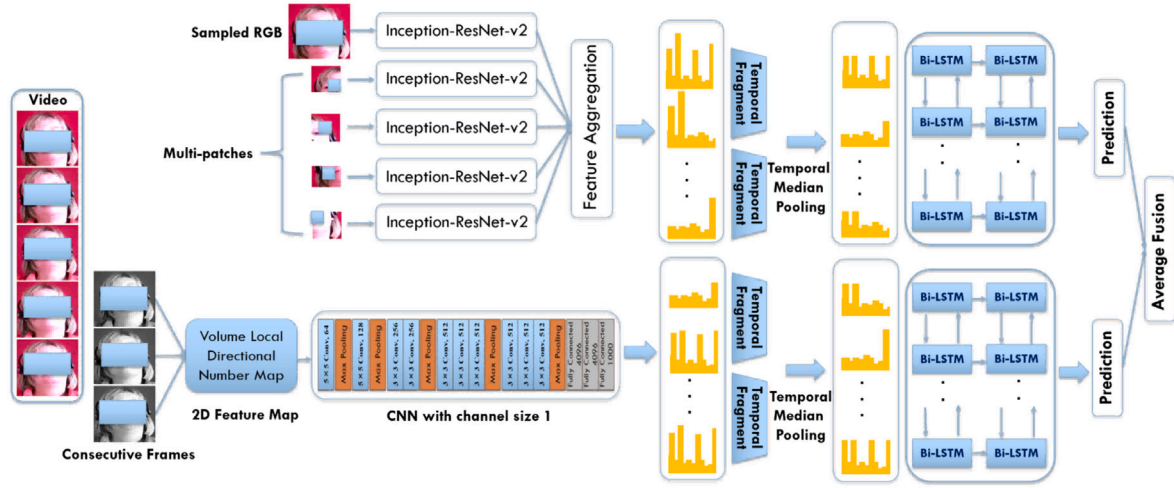


Fig. 9. The network using spatio-temporal features and multi-layer Bi-LSTM in this paper [46].

Chen et al. [50] propose a sequential fusion method leveraging facial appearance and dynamics to mine correlated and complementary depression patterns. Sequential fusion is achieved by feeding the mid-level features and predicted labels from the appearance stream into the dynamics stream, enabling an iterative refinement of the depression prediction.

He et al. [51] present an end-to-end trainable intelligent system. The system employs a 3D Convolutional Neural Network (3D-CNN) to extract local spatiotemporal features from video clips, capturing spatial and temporal depression-related patterns. A novel Spatiotemporal Feature Aggregation Module (STFAM) is introduced to refine these features, incorporating channel and spatial attention mechanisms alongside a 3D DEP-NetVLAD aggregation method to distil compact and discriminative representations. The system further utilizes Spatial Pyramid Pooling (SPP) to generate a high-level feature representation that is invariant to scale.

From 2019 to 2023, the research by de Melo et al. has significantly contributed to the advancement of DL-FEDR. In 2019, [52] presented a pioneering method to enhance depression detection by integrating Convolutional 3D (C3D) network predictions. These networks extract spatio-temporal features from global (full-face) and local (eye region) facial areas. The incorporation of 3D Global Average Pooling streamlines feature summarization, sidesteps the requirement for fully connected layers, and thus simplifies the model, reducing the complexity and the likelihood of overfitting. In 2020, [53] tackled the limitations of 2D CNNs in capturing dynamic spatio-temporal relationships by proposing a method that translates spatio-temporal dynamics into an image map via temporal pooling. This technique employs a two-stream model for the fusion of appearance and dynamic information at a late stage, surpassing contemporary methods in performance on the AVEC datasets. Also in 2020, [40] introduced the Multiscale Spatiotemporal Network (MSN), a 3D CNN architecture that encodes spatio-temporal relationships intrinsic to facial expressions for depression recognition. The MSN features parallel convolutional layers with diverse temporal depths and receptive field sizes, allowing it to encompass a wide array of spatiotemporal changes. Moving forward to 2023, [54] addressed the challenges of 3D CNNs, including fixed temporal depth and high computational demands, with the Maximization and Differentiation Network (MDN). The MDN employs a maximization block to detect subtle facial changes and a difference block to record abrupt variations, eschewing the need for 3D convolutions. The MDN module adopts a structure reminiscent of residuals, which can be seamlessly incorporated into various CNN architectures, offering an economical solution for depression detection. Building on the MDN's foundation, [55] developed the Decomposed Multiscale Spatiotemporal

Network (DMSN), which further refines the extraction of multiscale spatiotemporal features. The DMSN, composed of DMSN-A, DMSN-B, and DMSN-C blocks, is proficient in analyzing spatiotemporal features across varying temporal scales and spatial dimensions. This architecture has proven effective in both depression detection and pain estimation tasks while also minimizing the number of parameters required, setting a new standard for state-of-the-art models. The progressive development of these models reflects a steadfast effort to overcome previous limitations and enhance the precision of detecting and analyzing depression and pain from facial expressions. These innovations underscore the significance of capturing multiscale spatio-temporal information and hold promise for cost-effective, precise diagnostic applications in the healthcare sector.

Pan et al. [56] introduce the Spatial-Temporal Attention Depression Recognition Network (STA-DRN), engineered to capture essential global and local spatial-temporal information, thereby improving the precision of depression recognition from video data. The backbone of STA-DRN is a 3D residual network that integrates four bottleneck blocks, each composed of two bottleneck structures (see Fig. 10). A pivotal aspect of this approach is the novel Spatial-Temporal Attention (STA) module, which eschews the traditional $3 \times 3 \times 3$ convolution in favor of a more advanced mechanism. This module is divided into two sub-modules: a spatial attention module that generates the feature X_s and a temporal attention module that produces the feature X_t . Alongside these features, attention vectors S_t and S_s are created. The vectors are merged through an element-wise product, as shown in Eq. (1):

$$S_{st} = S_t \odot S_s \quad (1)$$

The output of the STA module, denoted as X_{fused} , is then calculated by applying the product of S_{st} and the sum of X_s and X_t , as outlined in Eq. (2):

$$X_{fused} = S_{st} \odot (X_s + X_t) \quad (2)$$

The STA-DRN is pre-trained using the UCF101 dataset [57] to establish temporal priors for video data and the CK+ dataset [58] for developing spatial priors in facial images. After pre-training, the model undergoes fine-tuning with the AVEC2013 and AVEC2014 databases, optimizing its performance for identifying signs of depression.

In this work [59], a lightweight Multi-scale Temporal Difference Attention Network (MTDAN) is proposed for automated video depression detection (see Fig. 11). The framework consists of two primary sub-branches: the Short-term Temporal Difference Attention Network (ST-TDAN) and the Long-term Temporal Difference Attention Network (LT-TDAN), which independently capture short-term and long-term facial dynamics. These networks are complemented by an Interactive

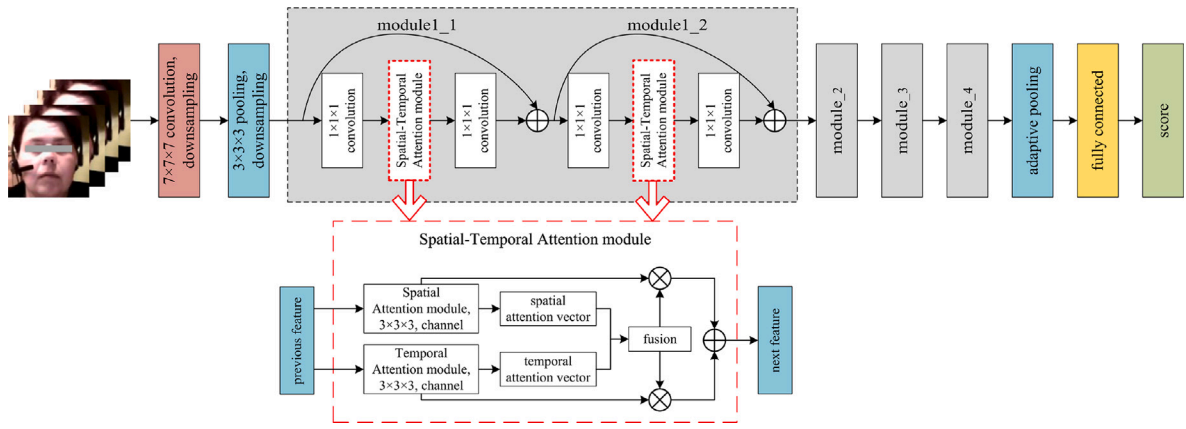


Fig. 10. The detailed architecture of STA-DRN [56].

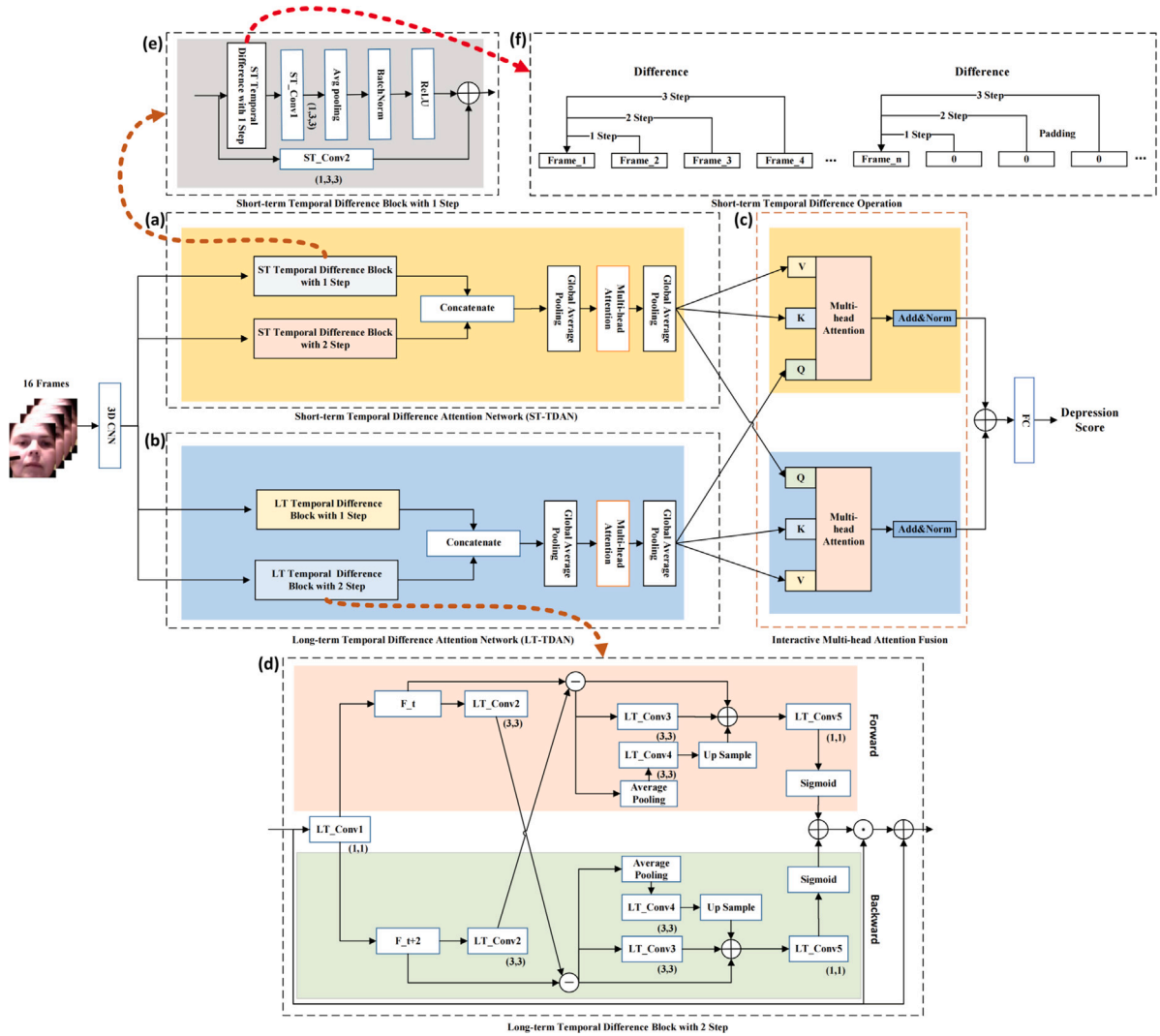


Fig. 11. The framework of MTDAN [59].

Multi-head Attention Fusion (IMHAF) strategy that seamlessly integrates the extracted features to predict depression scores. Notably, MTDAN achieves competitive performance on benchmark datasets such as AVEC2013 and AVEC2014 while maintaining significantly lower computational complexity than state-of-the-art methods.

Deep learning models for depression diagnosis face two challenges: learning long-term and short-term features effectively and merging them for video-based predictions. The Long Short-Term Cross-Attention-Aware Transformer (LSCAformer) addresses this with Long Short-Term Feature Extraction (LSTFE) and a Cross-Attention Transformer [62]. LSTFE captures long and short-term features, while the Cross-Attention

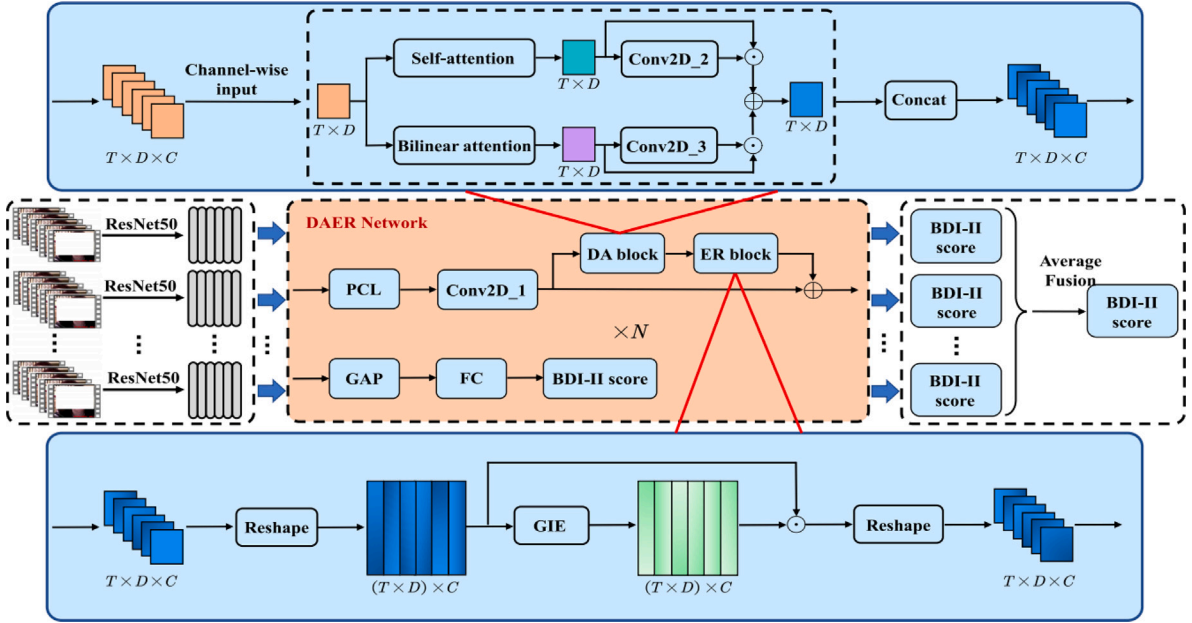


Fig. 12. Pipeline of the DAER network proposed in [60].

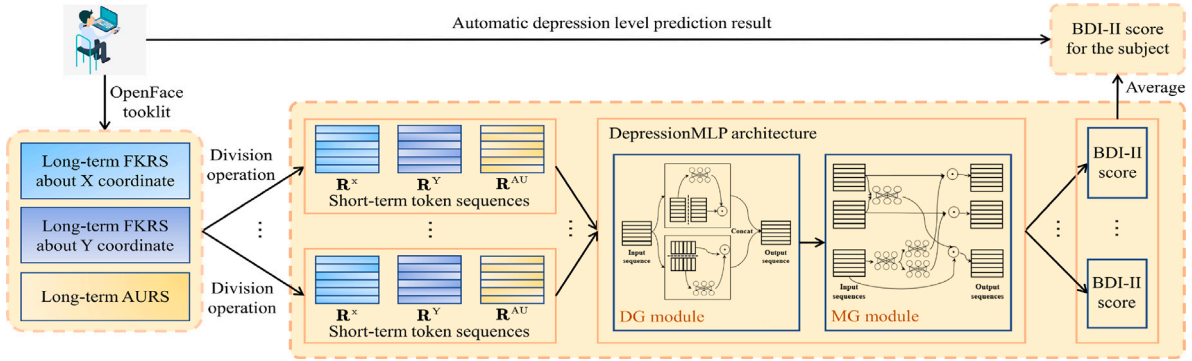


Fig. 13. Pipeline of DepressionMLP in [61].

Transformer uses Temporal-Directed Attention (TDA) to integrate these features and extract complementary temporal patterns, improving depression prediction accuracy.

Four recent papers [60,61,63,64] of Niu et al. have made notable contributions to DL-FEDR, each proposing innovative neural network architectures tailored to capture the subtleties of facial dynamics indicative of depression. [60] introduces the Dual Attention and Element Recalibration (DAER) network, which employs a dual attention mechanism to analyze the temporal changes in facial expressions and recalibrate elements within the tensor to represent depression cues better. This approach demonstrates a nuanced understanding of the importance of temporal dynamics and feature representation in depression analysis (refer to Fig. 12). [63] presents the Selective Element and Two Orders Vectorization (SE-TOV) network, which addresses the limitations of traditional convolutional neural networks by selectively choosing effective elements from multi-scale tensors and vectorizing high-dimensional data without pooling layers. This method ensures the preservation of detailed facial cues crucial for accurate depression severity diagnosis. [64] describes the Depressioner model, which incorporates a Graph Convolution Embedding (GCE) block to examine channel relationships and a Multi-Scale Vectorization (MSV) block to process temporal data. The model's ability to integrate channel advantages and extract multi-scale representations along the temporal dimension offers a comprehensive approach to depression-level prediction. [61] introduces the DepressionMLP architecture. This model

utilizes a novel Multi-Layer Perceptron (MLP) with gating mechanisms, including a Dual Gating module to maintain spatiotemporal attributes and a Mutual Guidance module to interact between Facial Keypoints Representation Sequences (FKRS) and Action Units Representation Sequences (AURS) (refer to Fig. 13). While all three models utilize the AVEC 2013 and AVEC 2014 databases for validation (the results refer to Table 2) and focus on enhancing the representation of facial cues, they differ in their architectural innovations. The DAER network's dual attention mechanism, the SE-TOV model's selective element approach, the Depressioner network's graph convolution and multi-scale vectorization, and the DepressionMLP model's Dual Gating and Mutual Guidance module each provide unique insights into the complex task of depression level prediction.

To summarize [65–67], here are several key points: (1) They discuss deep multi-modal frameworks that utilize facial and verbal cues. (2) Visual cues are highlighted as significant features within these frameworks. (3) [67] presents a framework that employs a Volume Local Directional Structural Pattern (VLDSP) descriptor to encode the structural aspects of local dynamic textures in video segments. This is combined with appearance features from the Inception-ResNet-V2 model. Besides, [65] introduces a Spatio-Temporal Attention (STA) network, which integrates spatial and temporal information, emphasizing frames relevant to depression detection. Additionally, [66] proposes an audio-visual attention network (AVA-DepressNet) with a privacy-preserving

Table 2

The experimental results in this table were conducted on the AVEC2013 and AVEC2014 datasets and are primarily based on visual information. The results of these methods are categorized and displayed according to spatial and spatiotemporal features.

Category	Methods	Year	AVEC2013		AVEC2014	
			MAE	RMSE	MAE	RMSE
Spatial	OpticalDR [39]	2024	7.53	8.48	7.89	8.82
	reducing noisy annotations [36]	2022	7.02	9.37	6.95	9.24
	IIS-DepNet [32]	2021	7.36	9.17	7.26	9.03
	C-DepressNet [38]	2022	6.89	8.43	6.57	8.26
	DLGA-CNN [33]	2020	6.59	8.39	6.51	8.30
	LQGDNet [35]	2021	6.38	8.20	6.08	7.84
	ResNet-50+Expectation loss [34]	2019	6.30	8.25	6.15	8.23
	MR_DepressNet [30]	2018	6.20	8.28	6.21	8.39
	DRR_DepressionNet [31]	2020	6.18	8.08	6.14	8.01
Spatial–Temporal	LKCT [37]	2023	/	/	5.48	7.41
	Appearance&Dynamics DCNNs [41]	2017	7.58	9.82	7.47	9.55
	RNN-C3D [43]	2018	7.37	9.28	7.22	9.20
	3D-CNN+2D-CNN&LSTM [65]	2020	7.32	8.97	6.43	8.60
	VLDN&Inception-ResNet-v2+Bi-LSTM&TMP [46]	2020	7.04	8.93	6.86	8.78
	IIS(3D-CNN+STFAM+SPP) [51]	2021	6.83	8.46	6.78	8.42
	C3D+Label Distribution&Metric Learning) [45]	2020	6.63	8.37	6.59	8.30
	Global&Local C3D+3D-GAP [52]	2019	6.40	8.26	6.59	8.31
	DAER [60]	2022	6.28	8.13	6.14	8.07
	MDN-152 [54]	2021	6.24	7.55	6.06	7.65
	Sequential Fusion of Appearance&Dynamics [50]	2021	/	/	6.16	8.13
	3D-CNN+EEP+GBDT+SVR [49]	2021	6.19	8.02	6.14	7.98
	Behavior Primitives+ANNs&CNNs [47]	2020	6.16	8.10	5.95	7.15
	MTDAN [59]	2023	6.14	8.08	6.35	7.93
	STA-DRN [56]	2023	6.15	7.98	6.00	7.75
	DMSN [55]	2023	6.14	7.66	5.69	7.50
	Depressor [64]	2022	6.12	7.49	6.01	7.56
	SE-TOV [63]	2022	6.09	7.42	5.87	7.39
	MSN [40]	2020	5.98	7.90	5.82	7.61
	VisEnc+DFNet+AMSTraining [66]	2023	5.97	7.26	5.99	7.30
	RGB&Encoded Dynamics+ResNet-50 [53]	2020	5.96	7.97	6.20	7.94
	Inception-ResNet-v2+VLDSP+Encoder&Decoder [67]	2022	5.90	7.32	5.75	6.98
	Depressformer [68]	2024	5.49	7.47	5.56	7.22
	LSCAformer [62]	2024	5.89	7.69	5.91	7.55
	DepressionMLP [61]	2024	5.43	7.49	5.63	7.27
	LMTformer [69]	2025	6.12	7.75	6.05	7.97

Table 3

The experimental results presented in this table were obtained using the D-vlog dataset and are primarily derived from multimodal information. The results in parentheses specifically pertain to those based solely on visual information. All the methods discussed here fall under the spatiotemporal category.

Year	Model	Precision	Recall	F1-Score
2022	Depression Detector [70]	65.40(60.92)	65.57(61.32)	63.50(56.38)
2023	TBOS [71]	65.40	64.7	65.0
2022	TAMFN [72]	66.02	66.50	65.82
2023	CAIINET [73]	66.57	66.98	66.56
2024	Depressformer [68]	–(64)	–(54)	–(59)
2023	DepMSTAT [74]	71.53	75.60	73.51
2023	STST [75]	72.50	77.67	75.00
2024	MDAVIF [76]	74.25	76.00	75.25

approach. This approach leverages facial landmarks and an adversarial multistage (AMS) training strategy, focusing on spatial–temporal relationships within the landmarks. (4) The results of their video-based unimodal experiments are referenced in Table 2, demonstrating excellent performance compared to other methods.

The aforementioned papers and network architectures have essentially conducted related experiments on the AVEC2013 and AVEC2014 datasets. With the release of the D-vlog dataset and considering the relatively small size of the AVEC datasets, researchers in the field of DL-FEDR have also begun to utilize the D-vlog dataset for research on depression recognition through facial expressions. For instance, the Depressformer model [68] integrates a Video Swin Transformer to capture deep feature representations and employs a Depression Fine-Grained Local Feature Extraction (DFLFE) module to focus on nuanced local features that are suggestive of depressive states. Additionally, the

depression channel attention fusion (DCAF) module is designed to combine the local and global features.

The Multi-Domain Acoustical-Visual Information Fusion (MDAVIF) network [76], also introduced in 2024, is an innovative approach to depression recognition from vlog data. It extracts multi-domain and multi-modal features from facial image sequences and aligned audio tracks. An Adaptive Feature Interaction (AFI) module is integrated to enhance the correlation between visual and acoustical data. The model also incorporates two autoencoders to retain information and prevent overfitting(refer to Fig. 15). The baseline model [70], termed Depression Detector, uses Transformer encoders to process the sequences of acoustic and visual features, and employs a cross-attention mechanism [77] to fuse these modalities effectively.

The TAMFN model [72] employs a time-aware attention mechanism to effectively mine and fuse features from multiple modalities, enhancing depression detection from non-verbal cues in vlogs. The STST method [75] integrates spatio-temporal data modules into a transformer encoder, capturing both temporal dynamics and spatial relationships, which improves detection accuracy(refer to Fig. 16). CAIINET [73] leverages contextual attention and information interaction to capture critical temporal information and effectively fuse multi-modal data. Lastly, the DepMSTAT framework [74] utilizes a Spatial–Temporal Attentional Transformer (STAT) to integrate spatio-temporal features from social media, achieving high accuracy in depression detection. These approaches collectively underscore the significance of spatio-temporal features and advanced fusion techniques in enhancing the effectiveness of depression detection systems.

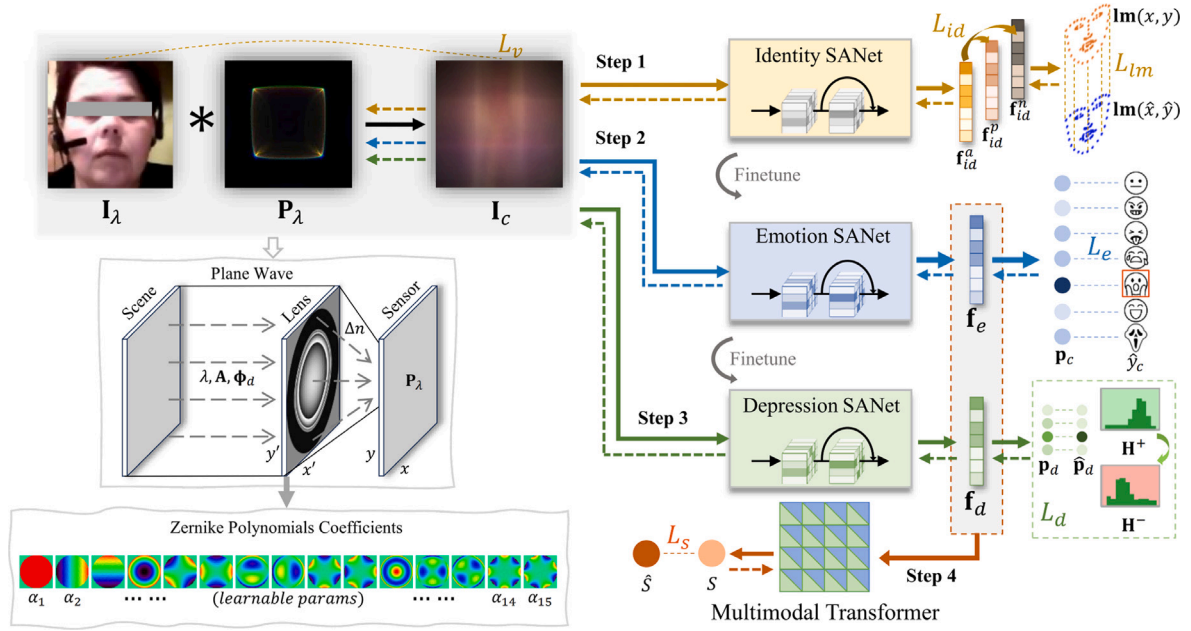


Fig. 14. Architecture of OpticalDR [39].

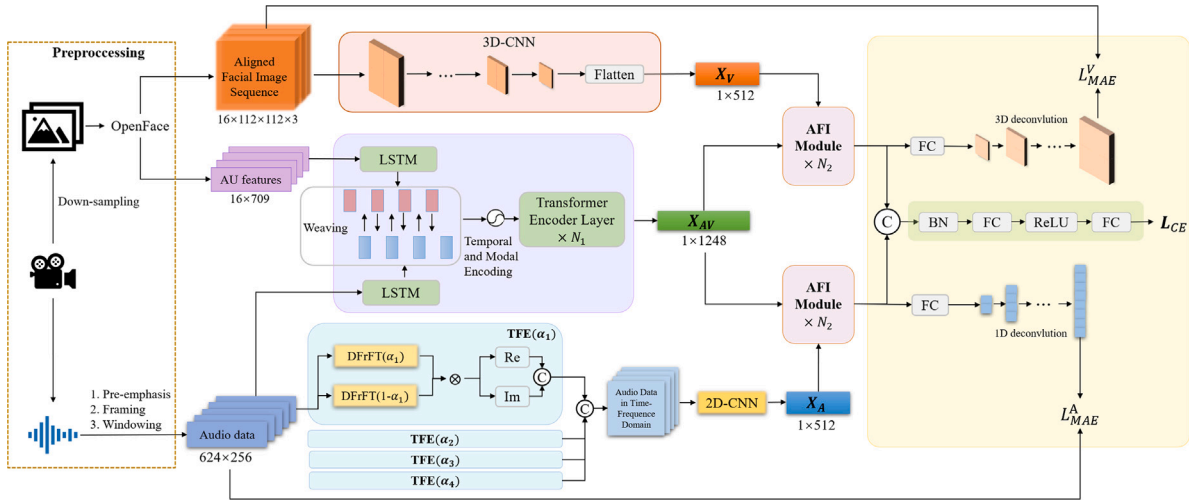


Fig. 15. The framework of MDAVIF [76].

3.2.1. Assessment of spatio-temporal features for DL-FEDR: strengths and limitations

Spatio-temporal feature-based video analysis methods provide significant advantages for depression recognition. By combining both spatial and temporal features, these methods enable the model to capture a more comprehensive understanding of behavioral patterns in videos, which leads to more accurate predictions of depression levels. For instance, the 3D Convolutional Neural Network (C3D) [43] can simultaneously learn both spatial and temporal features from video data, effectively capturing dynamic changes in facial expressions, which are essential for detecting emotional fluctuations over time.

However, processing video data requires substantial computational resources, particularly when handling long time-series data, which significantly increases the computational complexity [40]. For example, while the MDN [54] architecture is effective in capturing complex patterns, its inherent complexity – especially the combination of maximization and differential blocks – can reduce the interpretability of the model. Additionally, practical deployment of such models often requires extensive tuning and optimization, which can be time-consuming and challenging, particularly in resource-constrained environments.

3.3. Discussion

As we delve into the intricacies of DL-FEDR methods, this discussion will explore five critical dimensions: metrics, interpretability, ethical privacy, applicability and technology.

3.3.1. Metrics results

When training and testing depression recognition models on the AVEC 2013 and AVEC 2014 datasets, common evaluation metrics are the Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). As outlined in Eqs. (3) and (4), MAE and RMSE mean the average difference between the actual and predicted values in all examples, with smaller values indicating better model prediction performance.

$$MAE = \frac{1}{M} \sum_{i=0}^{M-1} |x_i - \hat{x}_i| \quad (3)$$

$$RMSE = \sqrt{\frac{1}{M} \sum_{i=0}^{M-1} (x_i - \hat{x}_i)^2} \quad (4)$$

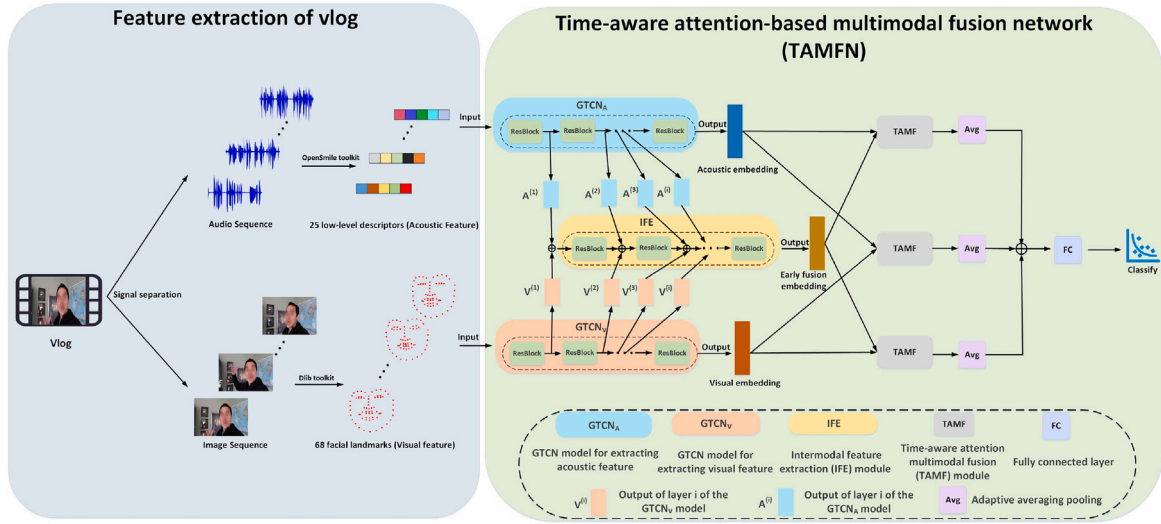


Fig. 16. The pipeline of TAMFN [72].

where M represents the total number of video samples, x_i is the true depression score of the i th video sample, and $\hat{x}_i$ is the predicted depression score of the i th video sample.

Table 2 lists the experimental results of RMSE and MAE in AVEC2013 and AVEC2014 for the above methods to detect depression from videos. From Table 2, on AVEC2013, we can see that the DepressionMLP method, proposed by Niu et al. [61], yields the best performance with an MAE of 5.43, and the visual model by Pan et al. [66] with an RMSE of 7.26. In addition, the best performance on AVEC2014 is an MAE of 5.48 from LKCT, proposed by Zhang et al. [37], and an RMSE of 6.98 from the Spatial-Temporal Network (only video) by Uiddin et al. [67].

For the D-vlog dataset, the classification performance is primarily assessed using three metrics: precision, recall, and F1-score. In this context, true positive (TP) refers to samples with positive labels that are correctly identified as positive. Likewise, true negative (TN), false positive (FP), and false negative (FN) correspond to samples correctly identified as negative, incorrectly identified as positive, and incorrectly identified as negative, respectively. The formulas can be expressed as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6)$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (7)$$

Table 3 presents the experimental results for Precision, Recall, and F1-Score in D-vlog. From Table 3, it is evident that MDAVIF [76] demonstrates the best overall performance, achieving the precision of 74.25% and the F1-score of 75.25%. Additionally, the highest recall of 77.67% is achieved by STST [75].

The generalization capabilities of various models across different datasets and tasks have been extensively analyzed, revealing that models like LKCT [37], MDN-152 [54], AVA-DepressNet [66], SE-TOV [63], DepressionMLP [61], and Depressformer [68] exhibit strong adaptability and robustness. These models, particularly those utilizing advanced neural network architectures, consistently outperform traditional hand-crafted feature-based approaches. For instance, the LKCT and AVA-DepressNet models demonstrate high accuracy and effective cross-dataset performance, while the SE-TOV network shows strong generalization across different facial expressions and datasets. The Depressformer model further exemplifies these capabilities, achieving competitive results on the AVEC2013, AVEC2014, and D-vlog datasets,

highlighting its ability to adapt to varied depression scales and facial dynamics.

Overall, the integration of sophisticated network structures, such as those found in Depressformer and AVA-DepressNet, provides enhanced generalization abilities, making these models reliable for diverse and complex depression recognition tasks across multiple contexts.

3.3.2. Interpretability analysis

Visualization techniques such as CAM [78], Grad-CAM [79], and Depression Activation Maps (DAM) [30] are vital tools for understanding and interpreting deep learning models used in depression detection from facial analysis. These methods help to identify which facial regions, like the eyes and mouth, are most relevant to the model's predictions [34,40]. By generating heatmaps, these visualizations reveal the areas that the models focus on when assessing depression severity, offering insights into how different facial expressions correspond to varying levels of depression [37,38,56,68]. DAM, specifically adapted for regression tasks, highlights facial regions that are crucial for estimating depression severity, making it a specialized tool beyond traditional classification-focused CAM methods [30].

Additionally, by integrating spatial and temporal information, these tools help refine model architectures, ensuring that both short-term and long-term facial dynamics are effectively captured [59]. Finally, visualizations not only enhance model accuracy but also provide a means to compare new models with existing state-of-the-art techniques [54]. By highlighting how new models capture relevant information more efficiently, these tools are crucial in demonstrating the advantages of newer approaches in terms of both accuracy and computational efficiency [39,55].

Overall, visualization methods are indispensable for advancing the field of automated depression detection, making models more interpretable, reliable, and effective.

3.3.3. Ethical privacy

The ethical use of DL-FEDR systems is paramount, especially concerning the privacy of individuals' facial expressions. As detailed in [39, 61,66], innovative strategies have been devised to protect privacy without hindering the models' ability to detect depression. For instance, AVA-DepressNet employs attention mechanisms to extract relevant features without raw facial imagery. Similarly, OpticalDR captures de-identified images, focusing on the essential characteristics for depression analysis (see Fig. 14). The DepressionMLP model processes sequences of facial keypoints and action units, ensuring privacy by not utilizing raw visual data.

For the development of the model, the conflict between privacy protection and recognition performance is a key challenge. When we aim to improve the model's depression recognition effectiveness, the need to protect users' facial privacy often results in insufficient data for model training, limiting the model's recognition performance. Currently, researchers can only share data within limited small circles and cannot publicly release facial video data, like the AVEC datasets. Although depression datasets may come from various sources, such as hospitals, psychologists, video surveillance, and social media platforms, the existence of data silos prevents data sharing across different institutions, making it unavailable for wider research use. This data hinder the development of depression recognition models because the vast potential data sources across various sectors cannot be fully utilized.

To address this issue, it is first necessary to improve relevant laws and regulations, as well as ethical usage guidelines for researchers. This would establish a safer and more standardized framework for data sharing and usage, ensuring legal and compliant use of data. Additionally, technical solutions for privacy protection, such as federated learning and differential privacy, are crucial. These methods allow for data collaboration and sharing across organizations while protecting user privacy. When these privacy protection frameworks mature and become practically applicable, it will be possible to train more generalized and effective models under privacy constraints.

Furthermore, when the size of raw data shared within a controlled and secure environment is sufficiently large, the effectiveness of depression recognition models can be maintained. Additionally, the use of extracted features such as facial landmarks, Action Units (AUs), and emotional data – particularly when combined with multimodal data fusion or when the datasets of extracted features, like D-vlog, are large enough – can enable the model to achieve satisfactory performance while still protecting privacy.

It is also essential to accelerate the development and adoption of advanced network architectures and related technologies, such as anonymization and de-identification techniques, which further enhance the model's ability to recognize depression without compromising privacy. By implementing these measures, it is possible to balance the needs for model recognition performance and privacy protection, promoting the further development of depression recognition technology.

3.3.4. Applicability analysis

When assessing the applicability of models, it is crucial to evaluate them based on the specific requirements of the application scenario. Different scenarios may have varying demands for recognition performance (including accuracy, generalization ability, and stability), real-time capability, interpretability, and privacy protection. Researchers in the DL-FEDR field are currently exploring methods that can enhance multiple aspects simultaneously.

Several studies have proposed innovative approaches to enhance the applicability of depression detection models. [37] introduces a deployable framework that secures sensitive information, using a novel two-stream deep network (LKCT) and a Purified Vision Transformer (PIT) for detailed image analysis. [38] presents a cloud-edge collaboration framework, with a shallow model (EdgeER) for quick emotion detection and a deep model (C-DepressNet) for precise depression analysis, showing excellent performance in accuracy and response time. [59] develops a lightweight Multi-scale Temporal Difference Attention Network (MTDAN) to model both short- and long-term facial behaviors for real-time video-based depression detection. [39] proposes the OpticalDR network, which uses a learnable lens for de-identified facial image recording, optimized for depression recognition. Moreover, [61] introduces DepressionMLP, an MLP architecture for depression level prediction based on facial keypoints and action units.

Additionally, the D-vlog dataset and related models highlight the importance of real-world applicability, focusing on data collected in natural environments rather than controlled laboratory settings. This

emphasis on practical, in-the-wild data makes the models more applicable to real-world scenarios where depression detection is critical.

Spatial and spatio-temporal feature methods are suited to different application scenarios. Spatio-temporal feature methods are better suited for video data sources, such as vlogs on social platforms and surveillance camera footage, where the sequential nature of video frames is crucial. On the other hand, spatial feature methods are more appropriate for image-based data sources, where there may not be a strong temporal relationship between multiple images, such as static photos posted on social media platforms like WeChat and Weibo, or frontal face photos stored by facial recognition systems at security checkpoints.

Image-based scenarios are relatively common and generally require less computational power than video processing, making them easier to handle and less demanding on hardware. However, because they lack the dynamic facial information provided by temporal sequences, image-based analysis may not be as effective in detecting depression and carries a higher risk of misinterpretation. Nonetheless, privacy protection is easier to achieve in image-based methods, and the data storage requirements are lower, making them necessary and valuable in practical applications.

In contrast, spatio-temporal feature methods for video analysis must account for dynamic changes over time as well as complex spatial features, which often results in larger and more complex models. Achieving high real-time performance in edge environments with limited hardware capabilities is challenging, and techniques like knowledge distillation are needed to lighten the models. Additionally, video data tends to be larger, with more potential for redundant or noisy data, requiring more complex models to ensure accurate recognition. Privacy protection is also more complex in video analysis, though techniques such as extracting action units (AUs) and facial landmarks can help anonymize facial data. However, this approach may not be as effective as analyzing the original data and could potentially overlook certain information related to depression.

The potential integration of DL-FEDR systems across various societal frameworks could significantly enhance mental health care. For instance, real-time depression detection could be integrated into telemedicine, offering more accessible mental health services, particularly for remote areas. In social platforms, continuous monitoring could provide early warnings for potential depression symptoms. Educational institutions could use these systems to identify and support students with mental health issues, while other industries could employ them to foster empathetic work environments. However, the deployment of such systems must be underpinned by robust privacy and security measures, ensuring user consent and trust.

3.3.5. Technological advancements

The evolution of DL-FEDR has been marked by a transition from simple 2D-CNN models to more complex architectures that integrate spatial and temporal data. Dual-stream and 3D-CNN architectures and attention mechanisms have significantly improved the models' ability to understand facial cues associated with depression. The introduction of transformer models has further refined feature extraction capabilities, focusing on identifying subtle depressive indicators.

The emphasis on multi-scale and multi-region analysis will likely continue, with models like the Decomposed Multi-scale Spatiotemporal Network (DMSN) demonstrating the benefits of a detailed approach [55]. The field is expected to see an influx of innovative approaches that will invigorate DL-FEDR and lead to more accurate and efficient systems.

3.3.6. Summary

In summary, this discussion has examined the multifaceted nature of DL-FEDR, from its reliance on precise metrics for evaluation to the ethical considerations surrounding privacy. It has also explored the potential applications of these systems across various sectors and the

technological advancements driving the field forward. As we prepare to conclude, it is clear that DL-FEDR holds promise for revolutionizing mental health diagnostics. The careful balance of technological innovation with ethical and privacy considerations will be crucial in realizing the full potential of these systems for the betterment of mental health care.

4. Challenges and directions

In the field of DL-FEDR, the evolution of this technology presents both significant challenges and promising future directions. This section aims to discuss the primary obstacles that researchers currently face in developing DL-FEDR systems, as well as the potential paths forward that could address these challenges and enhance the effectiveness and applicability of these systems.

The development of DL-FEDR faces several significant challenges. First, there is a limited availability of high-quality datasets, which is crucial for training effective models. The sensitive nature of depression, coupled with ethical concerns surrounding the collection of facial expression data, restricts the amount of data that can be ethically sourced and shared. Currently, raw facial data is deemed suitable for sharing only among a small group of researchers, rather than being made publicly available, to protect the privacy of individuals. On the other hand, derived data, such as facial landmarks, Action Units (AU), or emotion data, can be made publicly accessible as they pose a lower risk to privacy. Additionally, existing datasets often lack diversity in terms of demographics, which limits the generalizability of the models. Another challenge is the complexity and variability of facial expressions, which can be influenced by numerous factors such as lighting conditions, occlusions, and individual differences, making it difficult to develop robust and accurate models. Lastly, the ethical and privacy concerns associated with using personal facial data pose significant barriers to the widespread adoption of DL-FEDR technologies. Ensuring data security and protecting individuals' privacy while enabling accurate depression detection is a critical and ongoing challenge.

To overcome these challenges, collaboration across various sectors and stakeholders is essential. Effective research in this field requires the sharing of data under established ethical and privacy guidelines that are rigorously followed. One key future direction is the creation of more diverse and comprehensive datasets, which are ethically sourced and represent a wide range of demographics and real-world conditions. This would enhance the generalizability and effectiveness of DL-FEDR models. Another important area is the advancement of privacy-preserving techniques, such as federated learning, homomorphic encryption, and differential privacy, which would allow for the secure training of models without compromising individuals' privacy. Additionally, integrating multimodal data sources, such as audio, text, and physiological signals, alongside facial expressions, could improve the accuracy and robustness of depression detection systems. Furthermore, the development of real-time, edge-computing-based DL-FEDR systems could make these technologies more accessible and practical for use in everyday contexts, such as telemedicine and social media, where timely mental health assessments are crucial. The combined efforts of researchers, clinicians, technologists, and policymakers are necessary to ensure that these systems are developed responsibly, ethically, and effectively.

5. Conclusions

Our review has highlighted significant progress in applying deep learning techniques to analyze facial expressions for depression detection. A key insight is the importance of capturing multi-scale spatial-temporal data, which greatly improves depression detection accuracy. Advanced neural network models have played a crucial role in this advancement, offering enhanced precision in analyzing depressive states from facial cues. Furthermore, we have discussed pressing ethical

considerations, particularly regarding privacy. We emphasize the importance of implementing strategies that protect individuals' privacy while ensuring the continued effectiveness of depression detection algorithms. Our exploration has also revealed the potential for integrating depression detection systems into various societal structures, aiming to strengthen mental health care and take a proactive stance against depression.

In conclusion, we envision a future seamlessly integrating these systems across healthcare, social media, educational institutions, and other industrial sectors. However, successfully deploying such systems hinges on managing privacy concerns with the utmost respect and propriety. The potential applications of these systems are vast, offering promising avenues for improving mental health diagnostics and support.

CRediT authorship contribution statement

Xiaoming Cao: Writing – review & editing, Supervision, Methodology, Conceptualization. **Lingling Zhai:** Writing – review & editing, Writing – original draft. **Pengpeng Zhai:** Writing – review & editing. **Fangfei Li:** Writing – review & editing. **Tao He:** Writing – review & editing, Funding acquisition. **Lang He:** Writing – review & editing, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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